

Assignment D2-V8: Domestic Service Robot –

Cooling and Quality Degradation

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Project Overview

The project models a domestic service robot operating as an autonomous barista in a coffee shop environment. The robot (CafeBot) is equipped with a single manipulator (one-hand configuration), which introduces a physical constraint: it can hold only one beverage at a time. The objective is to demonstrate how delays in service lead to quality degradation and, ultimately, failure to serve the product.

Q1

PDDL model explanation

The model uses discrete state modeling to represent quality levels.

Quality states:

- Hot-fresh
- Warm-acceptable
- Cold-degraded

are modeled as predicates.

Conditional effect (when) is used to simulate quality loss over time, where each movement step forces a transition to a lower quality state. The robot can serve the dish only if it's hot-fresh or warm-acceptable. When it is cold-degraded it is not serveable. Each move action triggers a conditional effect that transitions the coffee state from hot-fresh to warm-acceptable, and eventually to cold-degraded.

Functions used in domain

The total-cost function acts as a heuristic metric to optimize service efficiency. By assigning specific weights to actions, with move operations costing 5 and quality-degradation transitions costing 10, we mathematically incentivize the robot to prioritize speed. This ensures that the robot selects the most efficient route and task sequence, reinforcing the 'delay leads to failure' constraint.

Plans and problems

Problem when timing is irrelevant

The goal is to serve black-coffee and espresso, the path is short in this problem. As the route is short the beverage can not get cold-degraded while the robot is serving coffee.

Generated plan

found plan:

found plan:

```
0.0: (pick-coffee cafebot black_coffee coffee_machine)
1.0: (move cafebot coffee_machine serving_area)
2.0: (check-temperature cafebot black_coffee)
3.0: (serve-coffee-warm cafebot black_coffee serving_area)
4.0: (move cafebot serving_area coffee_machine)
5.0: (pick-coffee cafebot espresso coffee_machine)
6.0: (move cafebot coffee_machine serving_area)
7.0: (check-temperature cafebot espresso)
8.0: (serve-coffee-warm cafebot espresso serving_area)
```

Problem when delays lead to failure

The goal is to serve espresso, the path is long in this problem, as it is elongated by some points. Long route makes it impossible for the robot to serve the coffee in an acceptable state.

Generated plan

Plan is not found, as the delay in this problem leads to failure

Discussion

This discussion highlights the importance of environmental topology in robotic planning. In real-world service robotics, task success is not only defined by reaching a destination but by maintaining product integrity under temporal constraints.

Q2

PDDL+ model explanation

Continuous Dynamics

We utilize :process to model the autonomous cooling of beverages (cooling) and the passage of time (timer-process), where (decrease (temperature ?d) (* #t 2.0)) allows for continuous state evolution.

Events

Threshold-based transitions are managed via :event. For example, end-move and end-pick trigger automatically once the action timer hits its threshold, representing physical reality more accurately than instantaneous discrete actions.

Autonomous Dynamics

The barista does not "cause" the coffee to cool; the world evolves independently of the planner's actions. Planning involves reasoning over continuous trajectories, ensuring the temperature remains above the threshold (40) upon arrival at the serving zone.

Functions used in domain

temperature ?d

A continuous fluent tracking the beverage's heat, which decreases autonomously over time.

action-timer ?r

A fluent that tracks the duration of ongoing robot actions, enabling the planner to trigger events once a task is completed.

total-cost

Used as a heuristic metric to penalize excessive service time, ensuring the plan remains economically efficient.

Plans and problems

Problem when timing is irrelevant

The goal is to serve two beverages. Given the initial temperatures and the robot's speed, the plan is feasible. The robot successfully coordinates pick-up and movement, completing the task while the coffee remains above the 40-degree quality threshold.

found plan:

0: (pick-coffee cafebot black_coffee coffee_machine)

0: -----waiting----- [2.0]

2.0: (move cafebot coffee_machine serving_area)

2.0: -----waiting----- [11.0]

11.0: (serve-coffee cafebot black_coffee serving_area)

11.0: (move cafebot serving_area coffee_machine)

11.0: -----waiting---- [16.0]

16.0: (pick-coffee cafebot espresso coffee_machine)

16.0: -----waiting---- [18.0]

18.0: (move cafebot coffee_machine serving_area)

18.0: -----waiting---- [23.0]

23.0: (serve-coffee cafebot espresso serving_area)

Problem when delays lead to failure

This scenario demonstrates "failure by delay." Due to the number of beverages (three instead of two), the timer-process forces the robot to spend more cumulative time picking and moving. The continuous cooling process causes the temperature of the third beverage to drop below 40 before the robot can reach the serving-zone. Plan is not found due to this.

Discussion

This model highlights the transition from classical symbolic planning to hybrid system control.

Planning as Trajectory Reasoning

Unlike the Q1 model, where degradation was forced by action counters, here the degradation is autonomous and continuous. This is a more faithful representation of physical reality, where inaction ("doing nothing") has a cost in terms of product quality.

System Constraints

The inability to serve the third coffee in the unsolvable problem highlights that in hybrid systems, task scheduling is fundamentally limited by the underlying dynamics of the environment. This model proves that identifying "failure by delay" in PDDL+ is a matter of predicting threshold violations in continuous state variables.